

Emmanuel Suárez Acevedo

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Education

Cornell University Ann S. Bowers College of Computing and Information Science
PhD in Computer Science

New York, NY
August 2024 – Present

University of Pennsylvania School of Engineering and Applied Science
MSE in Computer and Information Science

Philadelphia, PA
August 2022 – May 2024

University of Pennsylvania School of Engineering and Applied Science
BSE with Honors in Computer and Information Science

Philadelphia, PA
August 2015 – May 2019

Publications

Weighted NetKAT: A Programming Language For Quantitative Network Verification

PLDI 2026

Emmanuel Suárez Acevedo, Tiago Ferreira, Kevin Batz, Oliver Bøving, Nate Foster, and Alexandra Silva

Effects and Coeffects in Call-by-Push-Value

OOPSLA 2024

Cassia Torczon, Emmanuel Suárez Acevedo, Shubh Agrawal, Joey Velez-Ginorio, and Stephanie Weirich

Work Experience

Graduate Researcher

Cornell University

New York, NY
August 2024 – Present

- * Working on program logics for reasoning about concurrent, randomized programs
- * Developed a programming language for verifying quantitative network properties (e.g. latency, bandwidth)

Research Intern

University of Minnesota, Twin Cities

Minneapolis, MN
May 2024 – August 2024

- * Developed an extension to Martin-Löf type theory with strictly associative dependent sums and presented work at HoTT MURI 2024 Meeting

Graduate Researcher

University of Pennsylvania

Philadelphia, PA
August 2022 – April 2024

- * Developed type system for tracking effects and coeffects in call-by-push-value (with properties formalized in Rocq) and presented paper at OOPSLA 2024
- * Wrote paper in literate Agda (available on the arXiv) showcasing a proof that can be used for teaching the technique of logical relations with a proof assistant

Senior Software Engineer

Strava

San Francisco, CA
August 2019 – July 2022

- * Led the design and implementation of core system for modeling types of activities
- * Mentored junior engineers in microservice architecture and Scala programming
- * Used Apache Spark and Apache Kafka to build a geospatial data export tool for city planners
- * Built tools to verify and ensure correctness of internal route data
- * Developed a Scala microservice serving all static maps in product

Teaching Experience

Mentor

CIS 5520: Advanced Programming

University of Pennsylvania
Nov 2023 – Dec 2023

Teaching Assistant

CIS 5520: Advanced Programming

University of Pennsylvania
August 2022 – December 2022

Head Teaching Assistant

CIS 120: Programming Languages & Techniques I

University of Pennsylvania
August 2016 – May 2019

Service

POPL 2026 Artifact Evaluation Committee Member
Cornell Student-Applicant Support Program (Fall 2024)
Heights Philadelphia Summer School Teaching Assistant (2023)

Talks

Effects and Coeffects in Call-By-Push-Value	OOPSLA 2024
Strictly Associative Sigmas	HoTT MURI 2024 meeting

Awards

Dean's Excellence and Hopper-Dean Fellowship, Cornell University Ann S. Bowers College of Computing and Information Science (2024)
Outstanding Research Award, University of Pennsylvania School of Engineering and Applied Sciences (2024)
Graduate Research Fellowship, National Science Foundation (2024)
Dean's Master's Scholarship, University of Pennsylvania School of Engineering and Applied Sciences (2022)

Skills

Languages: Agda, OCaml, Rocq, Haskell, Scala, Lean, Java, Ruby, C/C++, Javascript
General: Git, L^AT_EX, Rails, Apache Spark, MySQL, React, Fluent in Spanish
